

Time Series Analysis

Seminar 10: State space models, Kalman filter and Markov switching



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **CBO**: Congressional Budget Office
- **EODHD**: EOD Historical Data (market-data provider)
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **HP**: Hodrick–Prescott (filter)
- **NBER**: National Bureau of Economic Research
- **OLS**: Ordinary Least Squares
- **SD**: Standard Deviation
- **SE**: Standard Error
- **SES**: Simple Exponential Smoothing
- **UC**: Unobserved Components (model)
- **US**: United States
- **VAT**: Value Added Tax

Today's Question and Seminar Structure

- The question of the seminar
  - **how do we estimate a quantity we never observe (a level, a trend, a regime) from noisy data, one period at a time?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the Kalman filter of a local level model for three periods; a missing value	Solved, Proposed	A1
A3, A4	steady state and exponential smoothing; state space forms and forecasts	Solved, Proposed	A3
A5, A6	durations of regimes; one step of the Hamilton filter	Solved, Proposed	A5
B1, B2	underlying Romanian inflation; the US output gap	Solved, Proposed	B1
B3, B4	recessions in US industrial production; BET volatility regimes	Solved, Proposed	B3
C1, C2	Romanian inflation regimes; what is wrong in an AI answer?	Proposed	B3, A1–A6

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	1996–2026
real GDP and potential GDP, US	FRED (GDPC1, GDPPOT)	quarterly	1947–2026
industrial production, US	FRED (INDPRO)	monthly	1960–2019
NBER recession indicator	FRED (USREC)	monthly	1960–2019
BET, S&P 500	EODHD	weekly, from daily closes	2000–2026

- Monthly inflation: $100\Delta \ln P_t$, seasonally adjusted (SA) by removing the monthly means (Chapter 4); CBO output gap: $100(\text{GDP}_t/\text{GDPPOT}_t - 1)$
- In the notebook: `kalman_by_hand`, `local_level_ml`, `uc_fit`, `hamilton_filter`, `MarkovRegression`; no account or key is needed

Prerequisites for Today (1/18): the state space form

- **State space form**: a hidden **state** α_t moves over time and is observed with noise Durbin and Koopman (2012)

$$\text{measurement: } y_t = Z\alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H)$$

$$\text{transition: } \alpha_{t+1} = T\alpha_t + R\eta_t, \quad \eta_t \sim N(0, Q)$$

- The system matrices
  - Z: links the state to the observation; T: moves the state one period ahead; R: maps the shocks η_t into the state
  - H, Q: the variances of the measurement noise ε_t and of the state shocks η_t; the two noises are independent
- Many models fit this form: ARMA, trends, cycles, regressions with time-varying coefficients; one algorithm (the Kalman filter) estimates them all

Prerequisites for Today (2/18): the local level model

- **Local level**: an observation = a level that wanders like a random walk + noise

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2); \qquad \mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2)$$

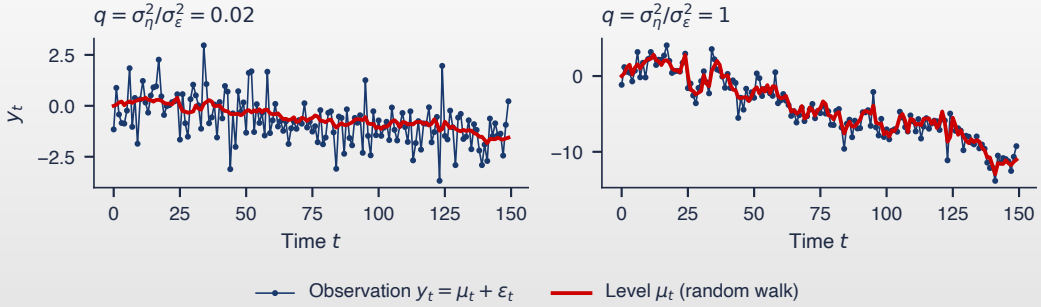
  - μ_t: the level (the state); $Z = T = R = 1$, $H = \sigma_\varepsilon^2$, $Q = \sigma_\eta^2$

- **Signal-to-noise ratio**

$$q = \frac{\sigma_\eta^2}{\sigma_\varepsilon^2}$$

  - q small: the level hardly moves, most changes of y_t are noise; q large: the level follows the data
  - $q = 0$: a constant mean; $\sigma_\varepsilon^2 = 0$: a pure random walk

Prerequisites for Today (3/18): the signal-to-noise ratio in a chart



- The same shocks, two values of q; left, $q = 0.02$: the level moves slowly and the observations scatter widely around it
- Right, $q = 1$: the level moves as much as the noise, so the data follow it closely

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Prerequisites for Today (4/18): three more models in state space form

- **Local linear trend**: a level and a slope, both random walks

$$\mu_{t+\mathbf{1}} = \mu_t + \beta_t + \eta_t, \qquad \beta_{t+\mathbf{1}} = \beta_t + \zeta_t, \qquad \alpha_t = \begin{pmatrix} \mu_t \\ \beta_t \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

  - β_t: the slope (growth per period); ζ_t: the shock to the slope; $Z = (1,\ 0)$

- **AR(2)** $y_t = \phi_{\mathbf{1}} y_{t-\mathbf{1}} + \phi_{\mathbf{2}} y_{t-\mathbf{2}} + \eta_t$: the state stacks the current and the previous value

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-\mathbf{1}} \end{pmatrix}, \quad T = \begin{pmatrix} \phi_{\mathbf{1}} & \phi_{\mathbf{2}} \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad Z = (1,\ 0), \quad H = 0$$

  - the second row of T only copies y_t into the next state; $H = 0$: y_t is observed exactly

- **Simple exponential smoothing** (SES, Chapter 0)

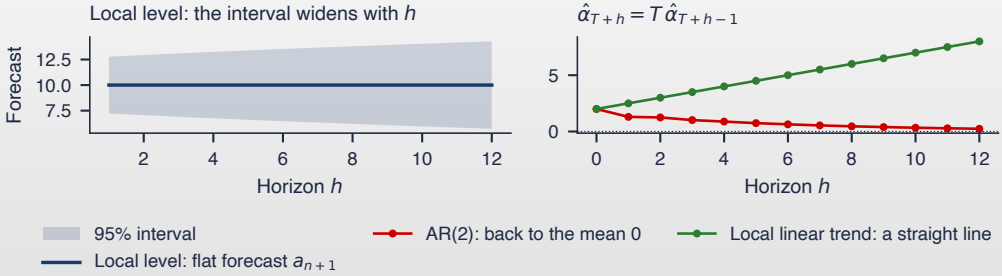
$$\ell_t = \alpha y_t + (1 - \alpha)\, \ell_{t-\mathbf{1}}, \qquad \hat{y}_{t+\mathbf{1}} = \ell_t$$

  - ℓ_t: the smoothed level; $\alpha \in (0, 1)$: the weight of the newest observation

Prerequisites for Today (5/18): forecasts from the state space form

- Forecasts iterate the transition from the last estimated state $\hat{\alpha}_T$ (the shocks are replaced by 0)

$$\hat{\alpha}_{T+h} = T\,\hat{\alpha}_{T+h-\mathbf{1}} = T^h \hat{\alpha}_T, \qquad \hat{y}_{T+h} = Z\,\hat{\alpha}_{T+h}$$



- Left: local level, all horizons equal $a_{n+\mathbf{1}}$; variance $P_{n+\mathbf{1}} + (h-1)\sigma_\eta^{\mathbf{2}} + \sigma_\varepsilon^{\mathbf{2}}$ grows with h. Right: AR(2) with $\phi_{\mathbf{1}} = 0.5$, $\phi_{\mathbf{2}} = 0.3$ and a trend with slope 0.5

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Prerequisites for Today (6/18): the Kalman filter of the local level model

- Before y_t arrives: a prediction a_t of the level μ_t and its variance P_t Kalman (1960)
  - start: a_1, P_1 given (a large P_1 when the start is unknown)
- **1. Surprise and its variance**

$$v_t = y_t - a_t, \qquad F_t = P_t + \sigma_\varepsilon^2$$

- **2. Update** with the **Kalman gain** K_t

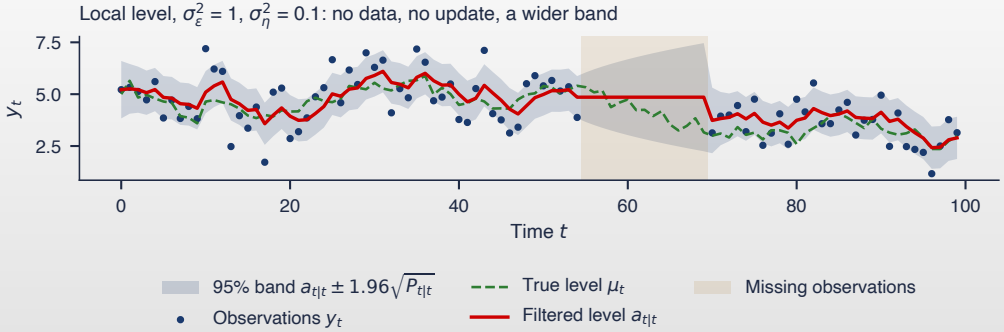
$$K_t = \frac{P_t}{F_t} \in (0,1), \qquad a_{t|t} = a_t + K_t\, v_t, \qquad P_{t|t} = P_t\,(1 - K_t)$$

  - K_t: the share of the surprise added to the level; uncertain level (P_t large) or precise data (σ_ε^2 small): K_t close to 1
- **3. Prediction** for the next period: the level is a random walk

$$a_{t+1} = a_{t|t}, \qquad P_{t+1} = P_{t|t} + \sigma_\eta^2$$

  - **missing** y_t: skip step 2, $a_{t|t} = a_t$, $P_{t|t} = P_t$; the variance grows by σ_η^2 each period without data

Prerequisites for Today (7/18): the Kalman filter in a chart



- The filtered level $a_{t|t}$ follows the true level with a delay; the band $\pm 1.96\sqrt{P_{t|t}}$ is its uncertainty
- Missing block: no update, $a_{t|t}$ stays flat and the band widens; the first observation after the gap gets a large gain

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Prerequisites for Today (8/18): the steady state and exponential smoothing

- After a few steps P_t and K_t stop changing: the **steady state** $\bar{P}$, $\bar{K}$

$$\bar{P} = \sigma_\varepsilon^2 \, \frac{q + \sqrt{q^2 + 4q}}{2}, \qquad \bar{K} = \frac{\bar{P}}{\bar{P} + \sigma_\varepsilon^2}$$

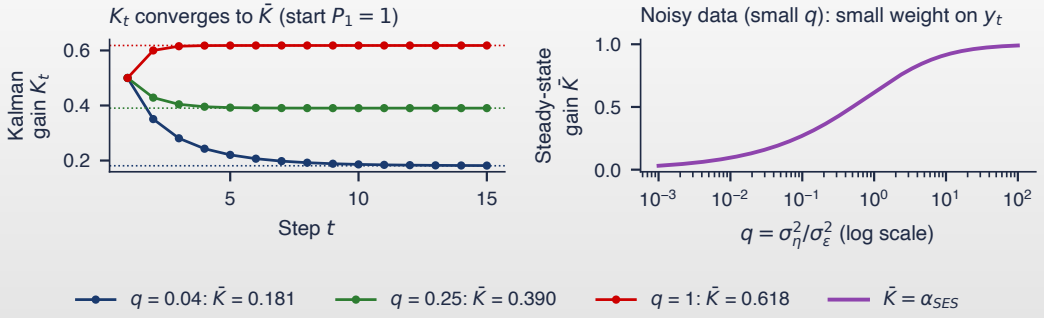
  - $\bar{P}$ is here the variance of the prediction a_t; it solves $\bar{P} = \bar{P}(1 - \bar{K}) + \sigma_\eta^2$

- In the steady state the filter **is** simple exponential smoothing Muth (1960)

$$a_{t+1} = a_t + \bar{K}(y_t - a_t) = \bar{K}\, y_t + (1 - \bar{K})\, a_t$$

  - the SES recursion with $\alpha = \bar{K}$ and $\ell_t = a_{t+1}$
  - inverse map, from an SES weight to the signal-to-noise ratio: $q = \alpha^2/(1 - \alpha)$

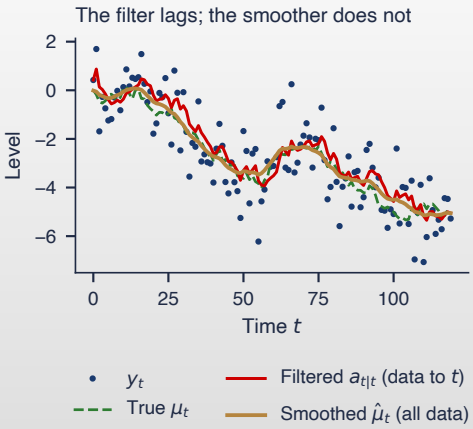
Prerequisites for Today (9/18): the Kalman gain in a chart



- Left: from $P_1 = 1$, K_t reaches $\bar{K}$ within a few steps; with $q = 1$, $\bar{K} = 0.618$ (the golden ratio minus 1)
- Right: $\bar{K}$, the equivalent SES weight, rises with q: the noisier the data (small q), the less weight on the newest observation

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Prerequisites for Today (10/18): likelihood, filtering and smoothing



- **Likelihood** by the prediction-error decomposition

$$\log L = -\frac{1}{2}\sum_{t=1}^{n}\left(\log 2\pi + \log F_t + \frac{v_t^2}{F_t}\right)$$

  - the filter gives v_t and F_t for given σ_ε^2, σ_η^2; we maximise $\log L$ over the two variances
  - large surprises v_t relative to their variance F_t lower the likelihood
- **Filtered** $a_{t|t}$: uses $y_1, \ldots, y_t$, what was known in real time
- **Smoothed** $\hat{\mu}_t$: uses the whole sample $y_1, \ldots, y_n$ (history) Rauch, Tung and Striebel (1965)
  - more precise, but revised when new data arrive; not available in real time

Prerequisites for Today (11/18): trend and cycle: the output gap

- **Output gap**: the cycle ψ_t in $y_t = 100 \log \text{GDP}_t = \mu_t + \psi_t$ (trend plus cycle, in % of the trend)
  - official benchmark in the United States: the CBO gap $100(\text{GDP}_t/\text{GDPPOT}_t - 1)$; CBO: Congressional Budget Office; GDPPOT: potential GDP
- **UC model** (unobserved components) Harvey (1989): a smooth trend plus an AR(2) cycle, estimated by the Kalman filter

$$\Delta^{\mathbf{2}}\mu_{t+\mathbf{1}} = \zeta_t, \qquad \psi_t = \phi_{\mathbf{1}}\psi_{t-\mathbf{1}} + \phi_{\mathbf{2}}\psi_{t-\mathbf{2}} + \kappa_t$$

  - $\Delta^{\mathbf{2}}\mu_{t+\mathbf{1}} = \mu_{t+\mathbf{1}} - 2\mu_t + \mu_{t-\mathbf{1}}$: the change of the slope; ζ_t, κ_t: independent shocks

Prerequisites for Today (12/18): two filters for the output gap

- **HP filter** Hodrick and Prescott (1997): the trend τ_t solves

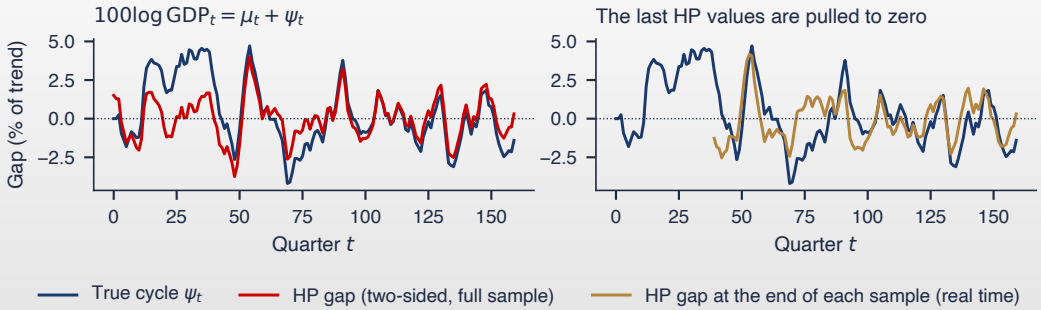
$$\min_{\tau} \sum_t (y_t - \tau_t)^2 + \lambda \sum_t (\Delta^2 \tau_t)^2, \qquad \lambda = 1600 \text{ for quarterly data}$$

  - fit against smoothness; two-sided: τ_t uses data after t, so the last values change when new quarters arrive
- **Hamilton filter** Hamilton (2018): the gap is the residual of an OLS regression two years ahead

$$y_{t+8} = b_0 + b_1 y_t + b_2 y_{t-1} + b_3 y_{t-2} + b_4 y_{t-3} + u_{t+8}, \qquad \text{gap}_{t+8} = \hat{u}_{t+8}$$

  - one-sided: uses only the past of each quarter

Prerequisites for Today (13/18): the HP filter in a chart



- Simulated quarterly 100 log GDP with a known AR(2) cycle; left: the full-sample HP gap follows the cycle (correlation 0.74)
- Right: the last HP value of each growing sample, what an analyst sees in real time: correlation 0.51, values pulled towards 0
- Correlation of two gaps a_t, b_t: $r = \sum(a_t - \bar{a})(b_t - \bar{b}) / \sqrt{\sum(a_t - \bar{a})^2 \sum(b_t - \bar{b})^2}$

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Prerequisites for Today (14/18): the Markov switching model

- **Markov switching** Hamilton (1989): the mean and the variance depend on a hidden **regime** $S_t \in \{1, 2\}$

$$y_t = \mu_{S_t} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma^2_{S_t})$$

  - μ_j, σ_j: the mean and the standard deviation in regime j (e.g. recession and expansion)
- The regime follows a **Markov chain**: tomorrow's regime depends only on today's

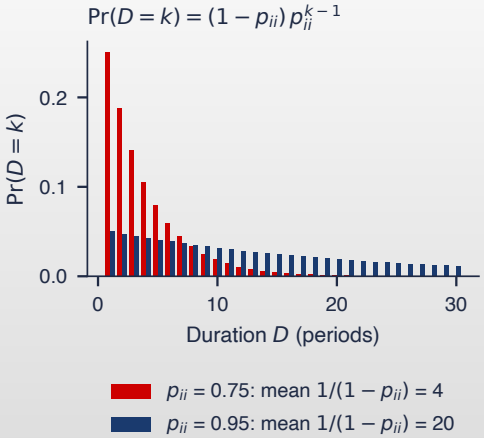
$$p_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \qquad \mathbf{P} = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}$$

  - each row of the transition matrix $\mathbf{P}$ sums to 1; h-step transition probabilities: the matrix power $\mathbf{P}^h$
- **Ergodic** (long-run) probability of regime 1

$$\pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \qquad \pi_2 = 1 - \pi_1$$

  - the share of periods spent in regime 1 over a long sample

Prerequisites for Today (15/18): regime durations



- Once in regime i, the chain stays with probability p_{ii} each period: the **duration** D is geometric

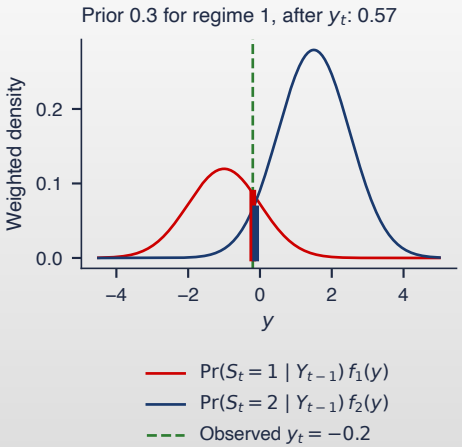
$$\Pr(D = k) = (1 - p_{ii})\, p_{ii}^{k-1}, \quad k = 1, 2, \ldots$$

  - stay $k - 1$ times, then leave once
- **Expected duration**

$$\mathbb{E}(D_i) = \sum_{k \geq 1} k\,(1 - p_{ii})\, p_{ii}^{k-1} = \frac{1}{1 - p_{ii}}$$

  - chart: $p_{ii} = 0.75$ gives 4 periods, $p_{ii} = 0.95$ gives 20
- Many short spells and a few long ones: the most likely duration is always 1

Prerequisites for Today (16/18): the Hamilton filter



- $\xi_{t-1} = \Pr(S_{t-1} = 1 \mid Y_{t-1})$; Y_{t-1}: the data up to $t-1$
  - **predict**:
    $\Pr(S_t = 1 \mid Y_{t-1}) = p_{11}\xi_{t-1} + (1 - p_{22})(1 - \xi_{t-1})$
- **Density** of y_t in regime j (Normal distribution)

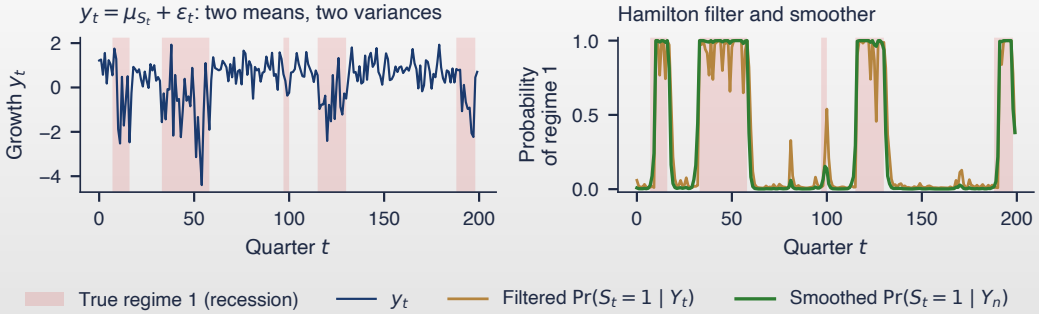
$$f_j(y_t) = \frac{1}{\sigma_j}\,\varphi\Big(\frac{y_t - \mu_j}{\sigma_j}\Big), \quad \varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}$$

- **Update** (Bayes rule): prior times density, normalised

$$\xi_t = \frac{\Pr(S_t = 1 \mid Y_{t-1})\, f_1(y_t)}{\sum_{j=1}^{2} \Pr(S_t = j \mid Y_{t-1})\, f_j(y_t)}$$

  - the denominator $p(y_t \mid Y_{t-1})$ is the likelihood contribution of period t
  - chart: prior 0.3; y_t is closer to μ_1, so $\xi_t = 0.57$

Prerequisites for Today (17/18): filtered and smoothed regime probabilities



- Simulated: $p_{11} = 0.90$, $p_{22} = 0.97$; regime 1: low mean, high variance; shaded: the true regime 1
- **Filtered** $\Pr(S_t = 1 \mid Y_t)$: real time, noisier; **smoothed** $\Pr(S_t = 1 \mid Y_n)$: uses later data, cleaner; classify a period in regime 1 if the probability exceeds 0.5

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Prerequisites for Today (18/18): regimes in practice: measures and checks

- **Concordance** with a reference dating (e.g. NBER recessions; NBER: National Bureau of Economic Research)
  - agreement: the share of periods classified like the reference; hit rate: the share of reference recessions found
  - false alarms: the share of reference expansions classified as recession
- **Annualised volatility** of regime j from weekly returns

$$\sigma_j^{year} = \sqrt{52}\,\sigma_j$$

  - 52 weeks a year; compare with the GARCH(1,1) volatility σ_t of Chapter 5
  - regime-implied volatility: $\sqrt{\xi_t \sigma_{\mathbf{1}}^{\mathbf{2}} + (1-\xi_t)\sigma_{\mathbf{2}}^{\mathbf{2}}}$, with ξ_t the probability of regime 1
- **Number of regimes**: compare 2 and 3 regimes by BIC $= -2\ln L + k \ln n$
  - k: the number of parameters (means, variances, transition probabilities); the smallest BIC wins

A1: the Kalman filter for three periods [Solved]

Task

- Local level with $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 1$; prior $a_1 = 0$, $P_1 = 1$; data $y = (1, 3, 2)$.
- 1. For $t = 1, 2, 3$ compute F_t, K_t, v_t, $a_{t|t}$, $P_{t|t}$ and the next prediction.
- 2. Give the forecast of y_4 and its variance.
- 3. Compute the steady-state gain $\bar{K}$ for $q = 1$.
- Report: a table of the three steps, one forecast, $\bar{K}$.

Solution

- 1. $t = 1$: $F = 2$, $K = 0.5$, $v = 1$, $a_{1|1} = 0.5$, $P_{1|1} = 0.5$; $a_2 = 0.5$, $P_2 = 1.5$
- $t = 2$: $F = 2.5$, $K = 0.6$, $v = 2.5$, $a_{2|2} = 0.5 + 0.6 \cdot 2.5 = 2$, $P_{2|2} = 0.6$; $a_3 = 2$, $P_3 = 1.6$
- $t = 3$: $F = 2.6$, $K = 0.615$, $v = 0$, $a_{3|3} = 2$, $P_{3|3} = 0.62$
- 2. $\hat{y}_4 = a_4 = 2.00$; $P_4 = 1.62$; Var $= P_4 + \sigma_\varepsilon^2 = 1.62 + 1 = 2.62$
- 3. $\bar{P} = (1 + \sqrt{5})/2 = 1.618$ (the golden ratio), $\bar{K} = 0.618$; K_t is already 0.615 at $t = 3$

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A2: a missing observation [Proposed]

Solution discussed in class

Task

- Local level with $\sigma_\varepsilon^2 = 2$, $\sigma_\eta^2 = 0.5$; $a_1 = 5$, $P_1 = 2$; data $y = (6, \text{missing}, 4)$. Model: A1.
- 1. Run the filter for $t = 1, 2, 3$, skipping the update at $t = 2$.
- 2. Explain why K_3 equals K_1.
- Report: the three steps and one sentence.

A3: exponential smoothing is a steady-state Kalman filter [Solved]

Task

- A local level model and simple exponential smoothing (SES).
- 1. Show that the steady-state update $a_{t+\mathbf{1}} = a_t + \bar{K}(y_t - a_t)$ is SES.
- 2. Compute $\bar{K}$ for $q = 0.25$ ($q = 1$ is in A1).
- 3. An SES model has $\alpha = 0.3$. Find the q of the local level model behind it.
- Report: one line of algebra, one gain, one q.

Solution

- 1. $a_{t+\mathbf{1}} = \bar{K}y_t + (1 - \bar{K})a_t$: the SES recursion with $\alpha = \bar{K}$ and $\ell_t = a_{t+\mathbf{1}}$
- 2. $q = 0.25$: $\bar{P}/\sigma_\varepsilon^{\mathbf{2}} = 0.640$, $\bar{K} = 0.390$
- 3. From $\bar{P} = \bar{P}(1 - \bar{K}) + \sigma_\eta^{\mathbf{2}}$: $\sigma_\eta^{\mathbf{2}} = \bar{K}\bar{P}$ and $\bar{P} = \alpha\sigma_\varepsilon^{\mathbf{2}}/(1 - \alpha)$, so $q = \alpha^{\mathbf{2}}/(1 - \alpha) = 0.09/0.7 = 0.129$
- a small α means a level that moves little relative to the noise

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A4: state space forms and forecasts [Proposed]

Solution discussed in class

Task

- An AR(2) with $\phi_1 = 0.5$, $\phi_2 = 0.3$, $\sigma^2 = 1$; a local linear trend. Model: A1 (the prediction step).
- 1. Write Z, T, R, Q and H for the AR(2).
- 2. With the last state $(y_T, y_{T-1}) = (2, 1)$, compute the forecasts for $h = 1, 2, 3$ by $\hat{\alpha}_{T+h} = T\hat{\alpha}_{T+h-1}$.
- 3. Write Z and T of the local linear trend and forecast three steps from $\mu_T = 100$, $\beta_T = 2$.
- Report: two sets of matrices and six forecasts.

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A5: durations of recessions and expansions [Solved]

Task

- A two-regime model of GDP growth: regime 1 (recession) with $p_{11} = 0.75$, regime 2 (expansion) with $p_{22} = 0.95$.
- 1. Write the transition matrix and compute the expected durations.
- 2. Compute the ergodic probabilities.
- 3. Starting in a recession, compute the probability of a recession two quarters later.
- Report: two durations, two probabilities, one two-step probability.

Solution

- 1. $\mathbf{P} = \begin{pmatrix} 0.75 & 0.25 \\ 0.05 & 0.95 \end{pmatrix}$; durations $1/0.25 = 4$ and $1/0.05 = 20$ quarters
- 2. $\pi_1 = 0.05/(0.25 + 0.05) = 0.167$, $\pi_2 = 0.833$: one quarter in six is a recession quarter
- 3. $p_{11}^{(2)} = 0.75^2 + 0.25 \cdot 0.05 = 0.575$; the chance of an expansion is 0.425
- the geometric duration: many recessions are short, a few are long; the mean is 4 quarters

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A6: one step of the Hamilton filter [Proposed]

Solution discussed in class

Task

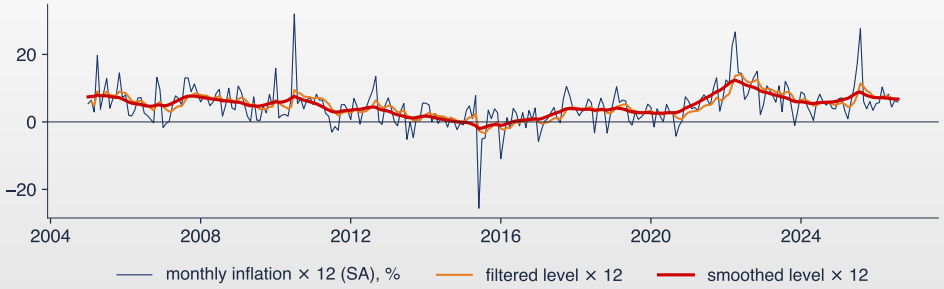
- The model of A5 with $\mu_1 = -0.5$, $\mu_2 = 1$, $\sigma = 1$; last quarter $\Pr(S_{t-1} = 1 \mid Y_{t-1}) = 0.2$; this quarter $y_t = -1$. Model: A5.
- 1. Compute the predicted probability $\Pr(S_t = 1 \mid Y_{t-1})$.
- 2. Compute the densities $f_1(y_t)$, $f_2(y_t)$ (standard Normal $\varphi(0.5) = 0.352$, $\varphi(2) = 0.054$) and the filtered probability.
- 3. Give the likelihood contribution and the predicted probability for next quarter.
- Report: four probabilities and one log-density.

B1: underlying Romanian inflation [Solved]

- **Question**: how fast does the underlying level of Romanian inflation move, and which exponential smoothing weight does the data choose?
- **Inputs**: monthly HICP inflation, seasonally adjusted, 2005–2026 ($T = 260$)
- **Tasks**
  1. Fit the local level model by maximum likelihood and report $\hat{\sigma}^2_\varepsilon$, $\hat{\sigma}^2_\eta$ and $\hat{q}$.
  2. Compute the steady-state gain $\bar{K}$ and compare it with the SES weight estimated directly.
  3. Plot the filtered and smoothed level ($\times 12$, in % per year) and report its peak and its last value.
  4. Interpretation: what does the size of $\hat{q}$ say about monthly inflation?
- **Report**: three variances, two weights, two levels, one sentence
- Notebook: section B1

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B1: solution [Solved]



- $\hat{\sigma}^2_\varepsilon = 0.142$, $\hat{\sigma}^2_\eta = 0.0057$, $\hat{q} = 0.040$; $\bar{K} = 0.181$; SES by least squares: $\hat{\alpha} = 0.175$
- Smoothed level $\times 12$: peak 12.3% (April 2022), low -2.0% (June 2015), last 6.8% (August 2026, SE 1.9); the raw series $\times 12$ has SD 5.6
- Interpretation: q is small: most monthly movements are noise (tax changes, food and energy prices); each month moves the underlying level by about 0.181 of the surprise

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B2: the US output gap [Proposed]

- **Question**: how close are three statistical output gaps to the official CBO gap, and which one would you use in real time?
- **Inputs**: US real GDP and the CBO potential GDP, quarterly, 1949–2026; model: B1
- **Tasks**
  1. Compute the UC cycle (smooth trend plus AR(2)), the HP gap ($\lambda = 1600$) and the Hamilton filter ($h = 8$, four lags) of 100 log GDP.
  2. Compute the CBO gap $100(\text{GDP}/\text{GDPPOT} - 1)$ and the correlation of each measure with it, including the filtered UC cycle.
  3. Report the standard deviations and the latest values.
  4. Interpretation: which measure would you use to judge the economy today?
- **Report**: five correlations, four standard deviations, four latest values, one sentence
- Notebook: section B2

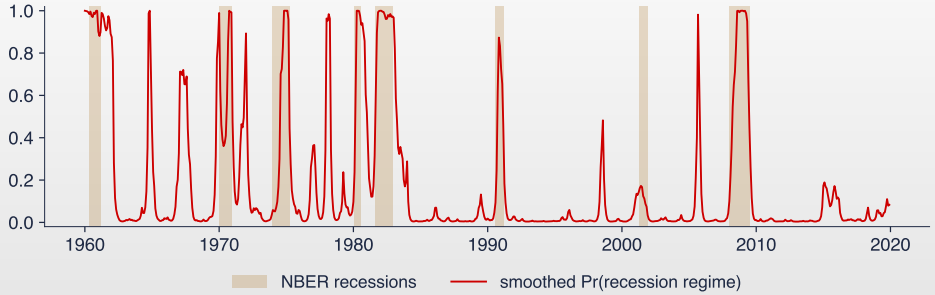
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B3: recessions in US industrial production [Solved]

- **Question**: does a two-regime model of industrial production find the NBER recessions?
- **Inputs**: monthly growth of US industrial production, 1960–2019 (93 NBER recession months)
- **Tasks**
  1. Fit a two-regime model with switching mean and variance (`MarkovRegression`).
  2. Report the regime means and standard deviations, p_{11}, p_{22} and the expected durations.
  3. Classify each month by the smoothed probability (> 0.5) and by the filtered probability, and compare with the NBER months.
  4. Interpretation: does the recession regime match the NBER recessions?
- **Report**: six parameters, two durations, two concordance tables, one sentence
- Notebook: section B3

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B3: solution [Solved]



- Recession regime: mean -0.16%, SD 1.30; expansion: mean 0.29%, SD 0.51; $\hat{p}_{11} = 0.874$, $\hat{p}_{22} = 0.970$; durations 8.0 and 33.2 months; ergodic share 19%
- Smoothed: 89% of months agree, 69% of NBER months found, 7% false alarms; filtered: 86%, 57%, 10%
- Interpretation: yes for the deep recessions; the regime is "low and volatile growth", so it also flags some slowdowns, and in real time it finds fewer recession months

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B4: volatility regimes of the BET [Proposed]

- **Question**: does the Bucharest market have calm and turbulent regimes, and are they the same as those of the S&P 500?
- **Inputs**: weekly log returns of the BET and of the S&P 500, 2000–2026; model: B3
- **Tasks**
  1. Fit a two-regime model with switching mean and variance to BET returns; report the annualised volatilities and durations.
  2. List the three years with the largest share of turbulent weeks.
  3. Fit a GARCH(1,1) (Chapter 5) and correlate its volatility with the regime-implied volatility.
  4. Interpretation: are the turbulent periods of the BET those of the S&P 500?
- **Report**: two volatilities, two durations, three years, two correlations, one sentence
- Notebook: section B4

C1: regimes of Romanian inflation [Proposed]

- **Question**: how many inflation regimes has Romania had since 1996, and when did the low-inflation regime begin?
- **Inputs**: monthly HICP inflation, not seasonally adjusted, 1996–2026; inflation targeting since August 2005; model: B3
- **Tasks**
  1. Fit models with two and three regimes (switching mean and variance) and compare them by BIC.
  2. For three regimes, report the means, durations and the date when the low regime becomes permanent.
  3. Propose one more check (seasonality, a regime with an AR term, out-of-sample forecasts).
  4. Interpretation: did inflation targeting start the low-inflation regime?
- **Report**: six parameters, a date, a check and a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant about state space and regime models. The answer:
- (a) For the Nile, q = 0.097, so the equivalent exponential smoothing weight is alpha = q = 0.097.
- (b) The Kalman gain is K = sigma2_eps / (P + sigma2_eps): the noisier the data, the more weight they get.
- (c) With p11 = 0.75, recessions last p11 / (1 - p11) = 3 quarters on average.
- (d) To call a recession in real time, use smoothed_marginal_probabilities from statsmodels.
- (e) The last value of the HP gap is a reliable estimate of today's output gap.
- (f) In the local level model the forecasts of all horizons are equal; only their variance grows with the horizon.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (Part A, notebook section C2).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Kalman filter: $K_t = P_t/(P_t + \sigma_\varepsilon^2)$, $a_{t|t} = a_t + K_t v_t$; a missing value means no update
- Exponential smoothing is a local level filter in steady state: $\alpha = \bar{K}$, $q = \alpha^2/(1-\alpha)$
- Romanian monthly inflation: small q, a slowly moving underlying level
- Markov switching: durations $1/(1-p_{ii})$; filtered probabilities for real time, smoothed ones for history
- An AI answer is a draft: check the gain, the duration formula, filtered against smoothed

After the seminar

- Lecture 10 develops each topic of today
  - the state space form, the filter and the smoother, the likelihood, trend and cycle, Markov switching
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation regimes in Romania and in the region
- Reading: Durbin and Koopman (2012), Ch. 2; Hamilton (1989); Hamilton (2018)

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